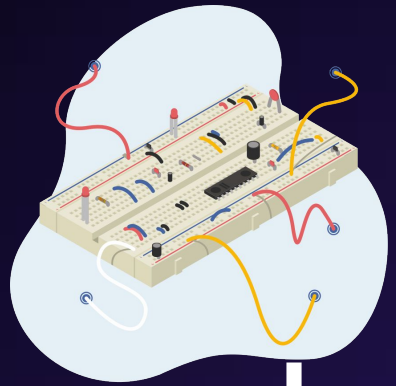
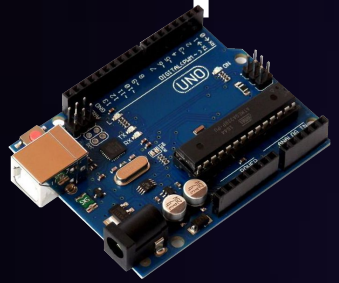
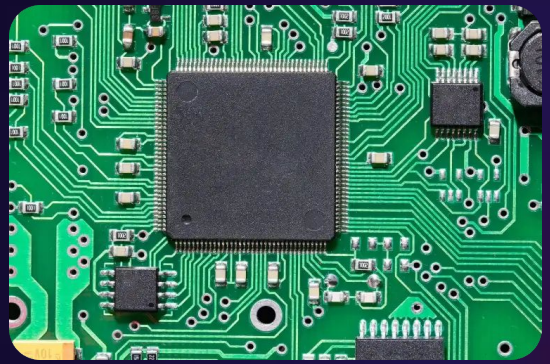
Architecture of electronic and computing machine and computing systems







Course Code: S002
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COURSE INTRODUCTION



COURSE STRUCTURE

Foundations → Systems → Integration → Assessment



COURSE ASSESSMENT

Final grade = 1 (40%) + 2 (60%)

1. Midterm assessment: **100 points**
2. Final project : **100 points**

MIDTERM ASSESSMENT BREAKDOWN

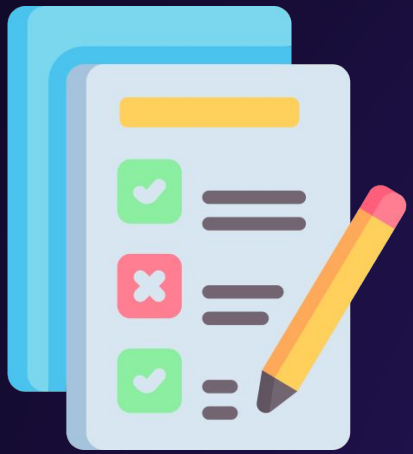
100 POINTS (100%)



- **10%** - performance
- **10%** - attendance
- **80%** - lab test (openbook)

FINAL ASSESSMENT

100 POINTS

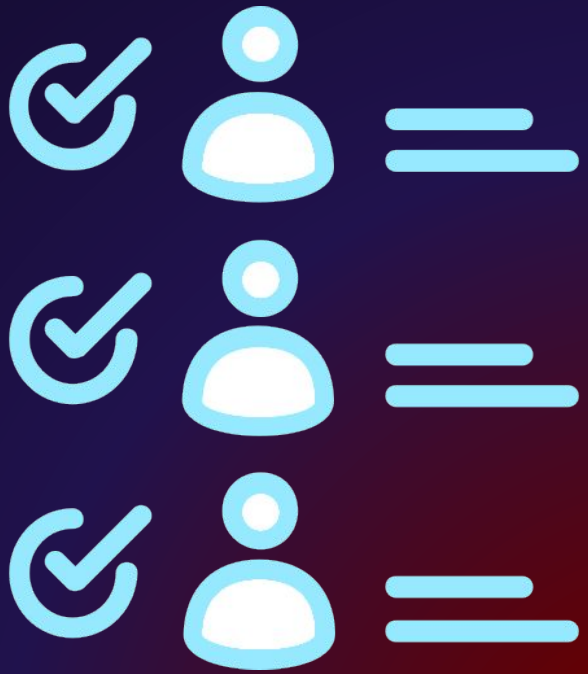


100% - final project

STUDENT RESPONSIBILITY

*** ATTENDANCE ***

You are expected to attend classes consistently, as course concepts build progressively



STUDENT RESPONSIBILITY

*** ACTIVE PARTICIPATION ***

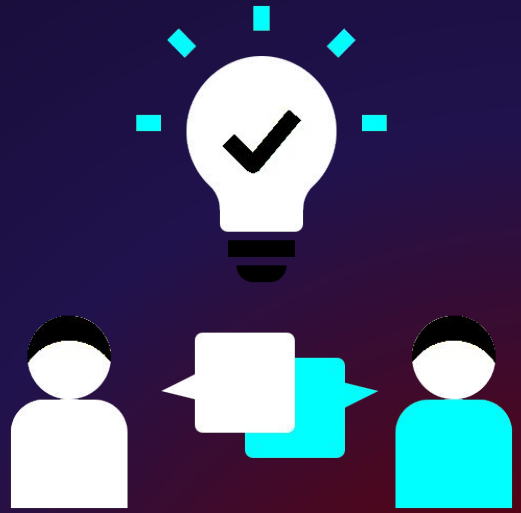
Engagement in discussions and in-class activities is essential for understanding the ideas and topics covered by this subject.



STUDENT RESPONSIBILITY

*** CONCEPTUAL UNDERSTANDING ***

The course emphasizes understanding how systems work and interact, rather than memorizing definitions or tools.



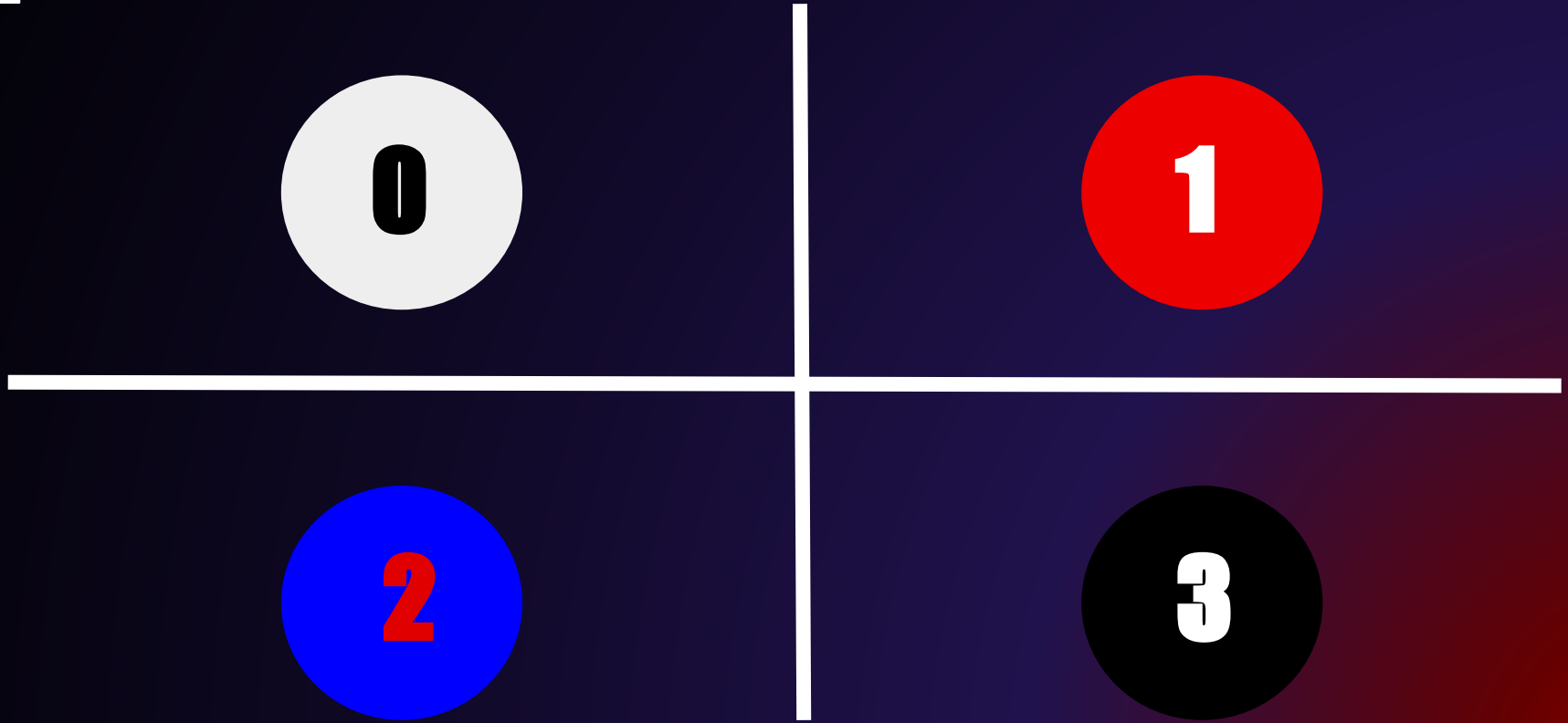
STUDENT RESPONSIBILITY

*** PROFESSIONAL BEHAVIOR ***

You are expected to communicate respectfully, meet deadlines, and follow academic integrity standards.



TEAMS



How do devices talk to each other

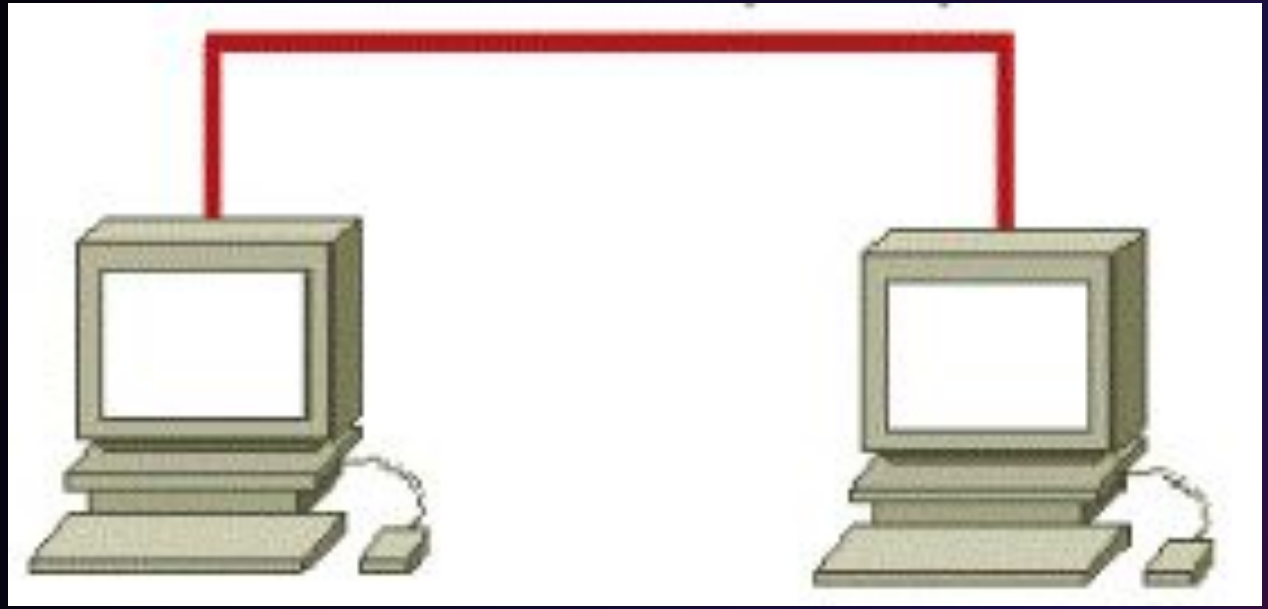




Agenda

- The problem models solve
- The OSI model (7 layers)
- The TCP/IP model (4 layers)
- Encapsulation (message travel)
- Introduction to Wireshark

Before standards



THE IDEA



1960’s

Proprietary networks

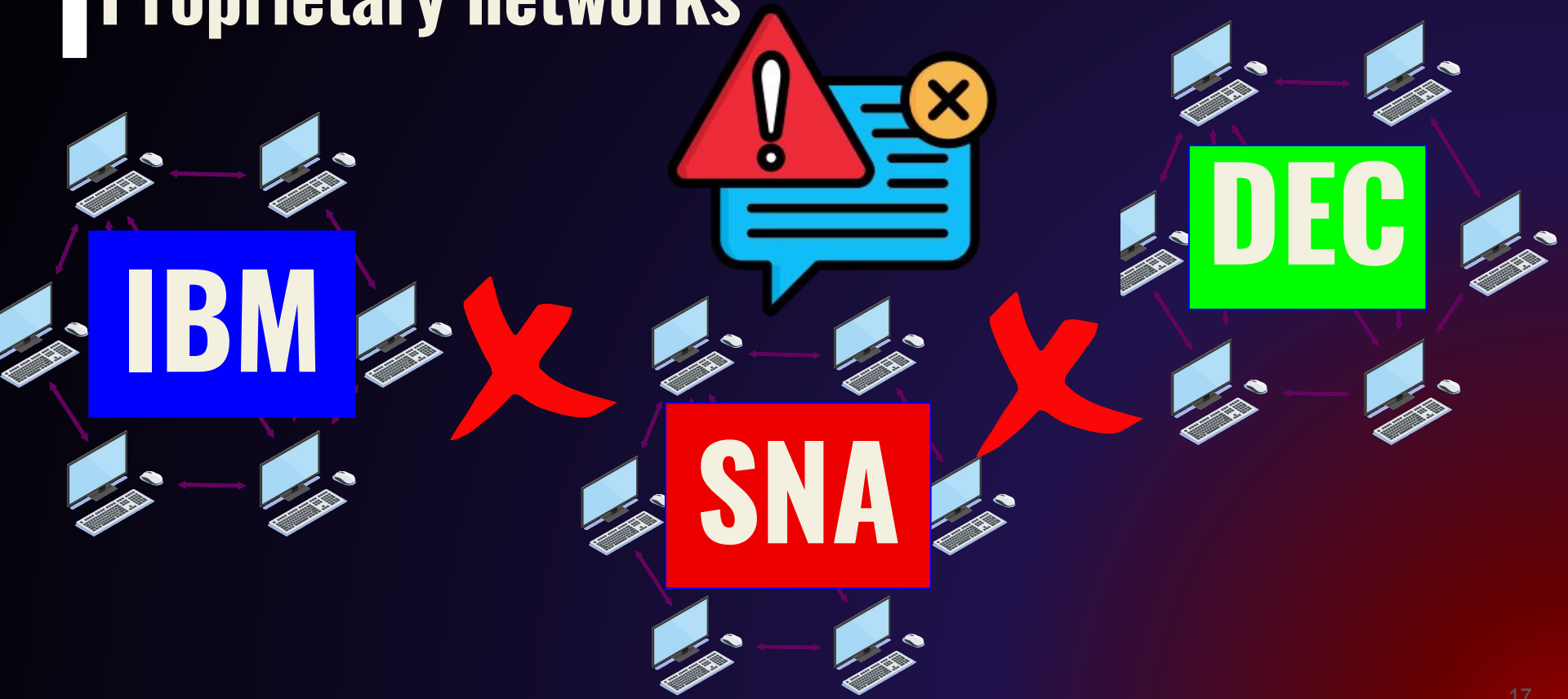
1970’s - 1980’s





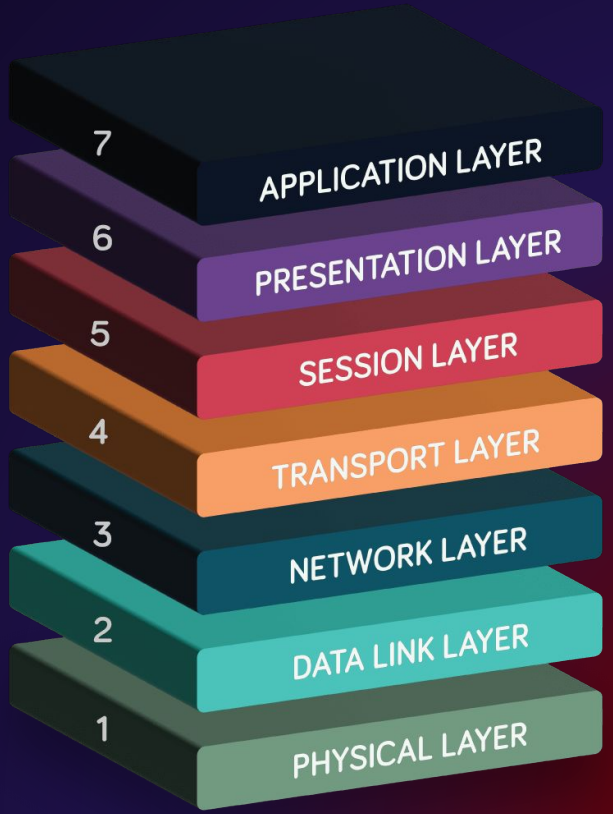


Proprietary networks



The Open Systems Interconnection model (OSI model)





Why bother with layers



Why bother with layers

Abstraction → each layer only needs to know its own job

Modularity → upgrade one layer's technology while every other layer stays untouched

Troubleshooting → isolate a fault to one layer instead of guessing

The envelope concept



The envelope concept

Write a letter → seal it in an envelope address it → post office adds it to a mail sack → truck carries the sack

At every step, the handler only needs to understand *their* part. Network layers work the same way: wrap, address, hand off without needing to understand the other layers.

The OSI model - Application layer (7)

The layer users interact with



This layer provides network services directly to applications used by people. Common protocols: HTTP / HTTPS, FTP, DNS, SMTP and MQTT.

The OSI model - Presentation layer (6)

Makes data readable and secure



This layer ensures data is presented in a format both devices understand. Data translation + encryption and decryption + compression and decompression.

The OSI model - Session layer (5)

Controls the conversation







It starts conversations, maintains active connections and ends sessions when communication finishes. Example: authentication (log into Gmail once and continue browsing multiple pages without logging in again)

The OSI model - Transport layer (4)

Responsible for delivery quality



+



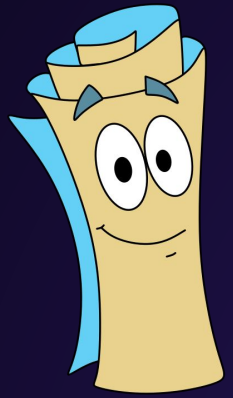
+



Ensures data reaches the correct application and arrives reliably when required. Segmentation of data + error detection + flow control + port management. Main protocols are TCP and UDP.

The OSI model - Network layer (3)

Responsible for finding the route



Determines where data should go across multiple networks. Logical addressing + routing + path selection. Routers with IPv4 or IPv6 and ICMP.

The OSI model - Data link layer (2)

Responsible for for local delivery

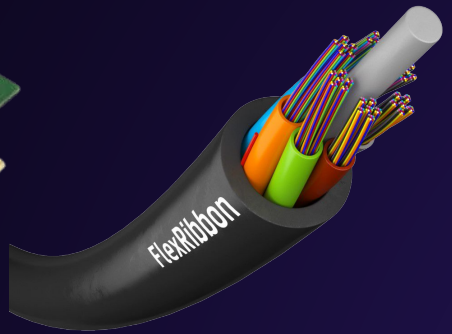


Transfers data between devices connected to the same local network. Uses MAC addresses + detects local transmission errors + creates frames.

The OSI model - Physical layer (1)

Moves the actual signals

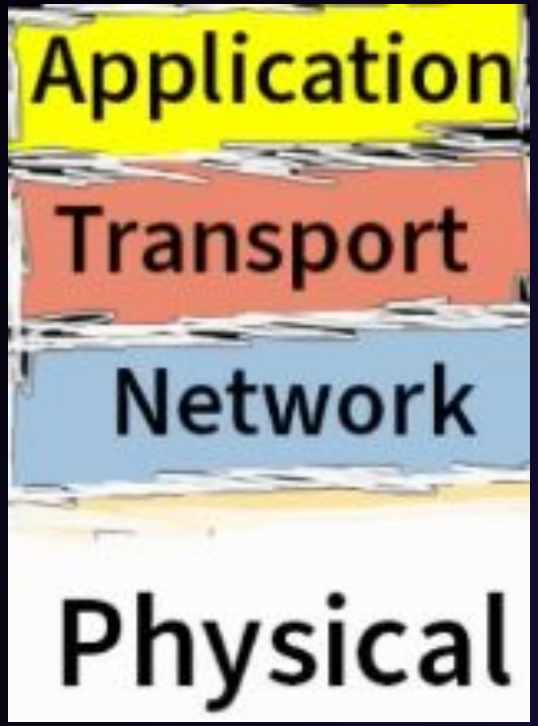






This layer transmits raw bits through physical media. Ethernet cables + Fiber optic cables + Wireless antennas + Network interface cards (NICs)

TCP/IP





What does it actually stand for



What does it actually stand for

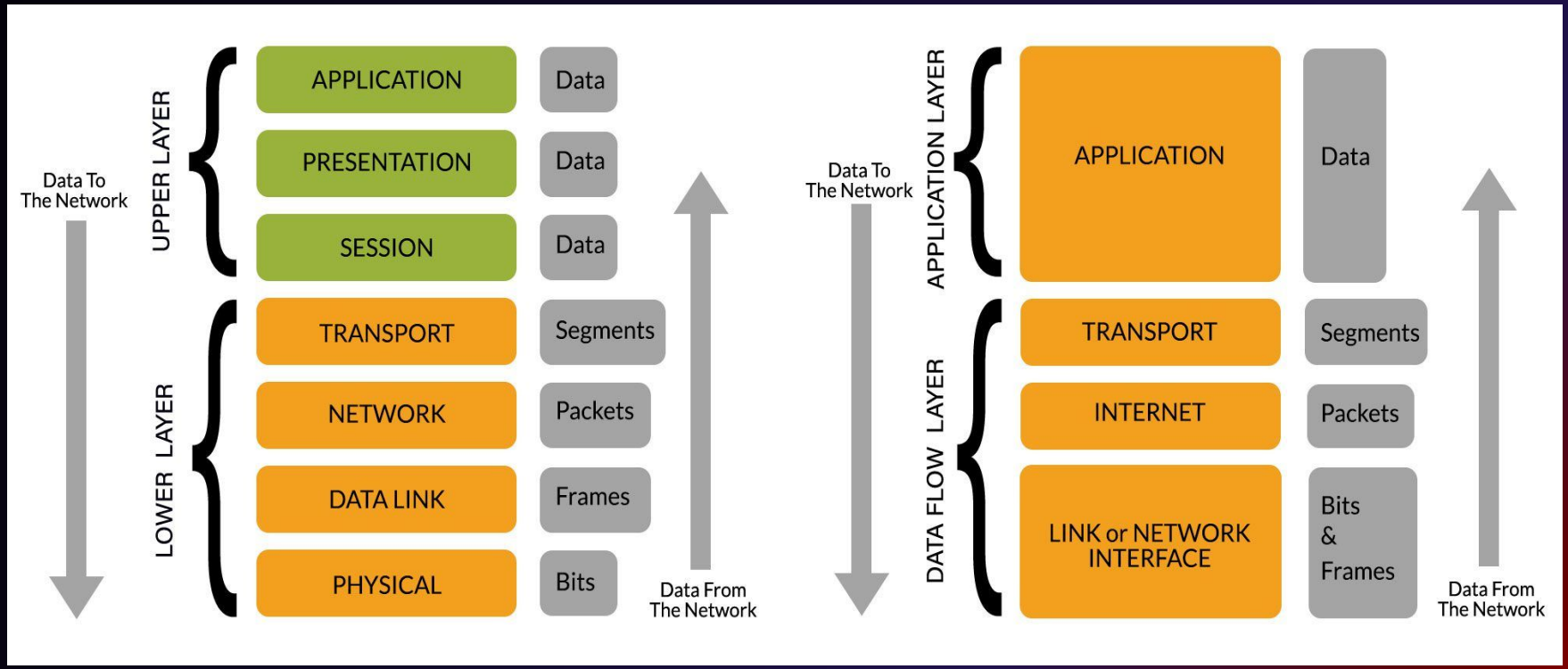
TCP/IP = Transmission Control Protocol / Internet Protocol

Think of it as a suite of protocols on which entire internet is working.

Why we learn both OSI and TCP/IP



OSI model vs TCP/IP model



Physical layer

TCP/IP model

Lowest layer in the stack which send raw bits (1s and 0s) over a real medium. Medium can be a cable, radio waves, or light. It d**oes not understand addresses or data, only signals**

Example: WiFi turns bits into radio waves and back into bits

Example: an Ethernet cable carries bits as voltage changes

If the cable is unplugged or WiFi is too weak, this is where the problem is

Physical layer

TCP/IP model



Link layer

TCP/IP model

This layer handles devices on the same local network, it groups bits into units called frames. Uses MAC addresses to identify devices nearby

Example: your laptop sends data to the classroom router using the router's MAC address

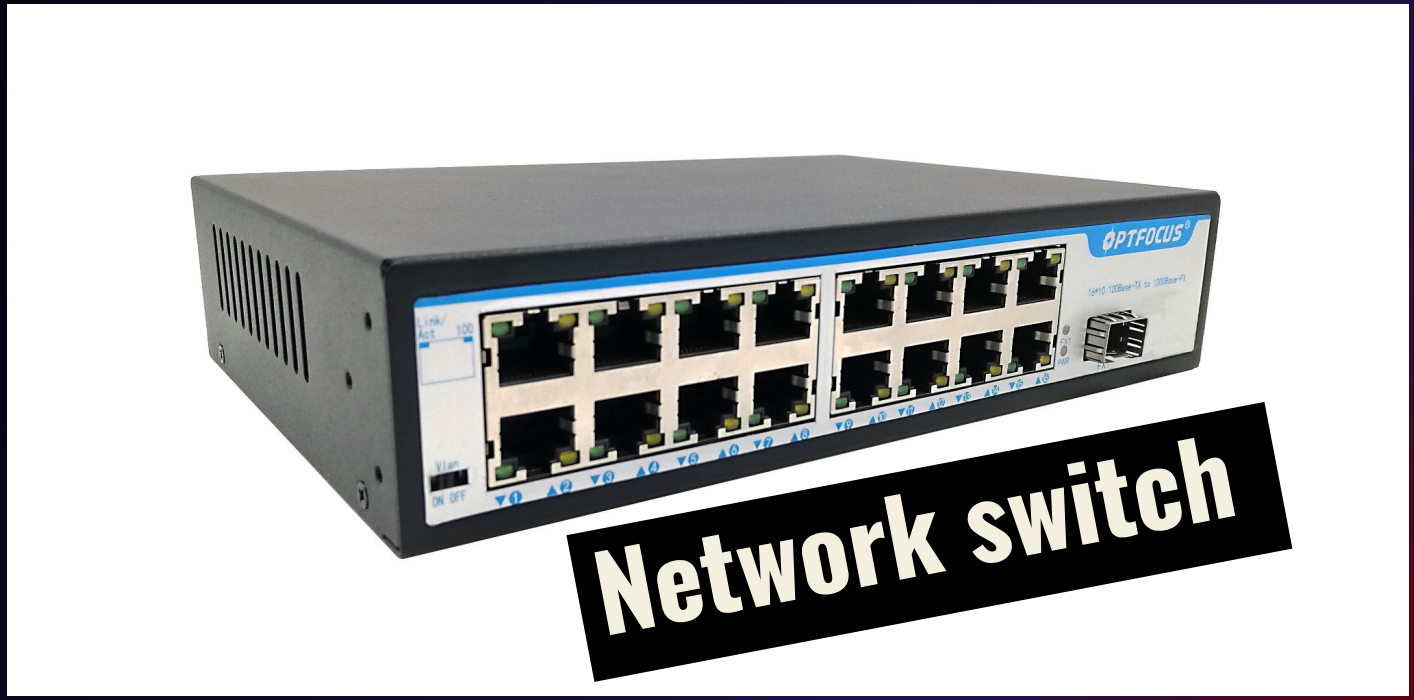
Protocols here: Ethernet, WiFi (802.11)

Switches use MAC addresses to send frames to the right device

Solves the problem of many devices sharing one network

Link layer

TCP/IP model



Network layer

TCP/IP model

This layer moves data between different networks, **NOT** nearby devices. Uses IP addresses to identify devices anywhere on the internet. Main protocol: IP. Routers read the IP address and pass the packet closer to its destination. This process is called routing

Example: a request from Bishkek passes through several routers to reach a server abroad

Related tools: ping uses ICMP, ARP turns an IP address into a MAC address

Network layer

TCP/IP model



Transport layer

TCP/IP model

Delivers data to the correct application, not just the correct device. It uses port numbers to identify which app the data belongs to

Example: port 80 is HTTP, port 443 is HTTPS

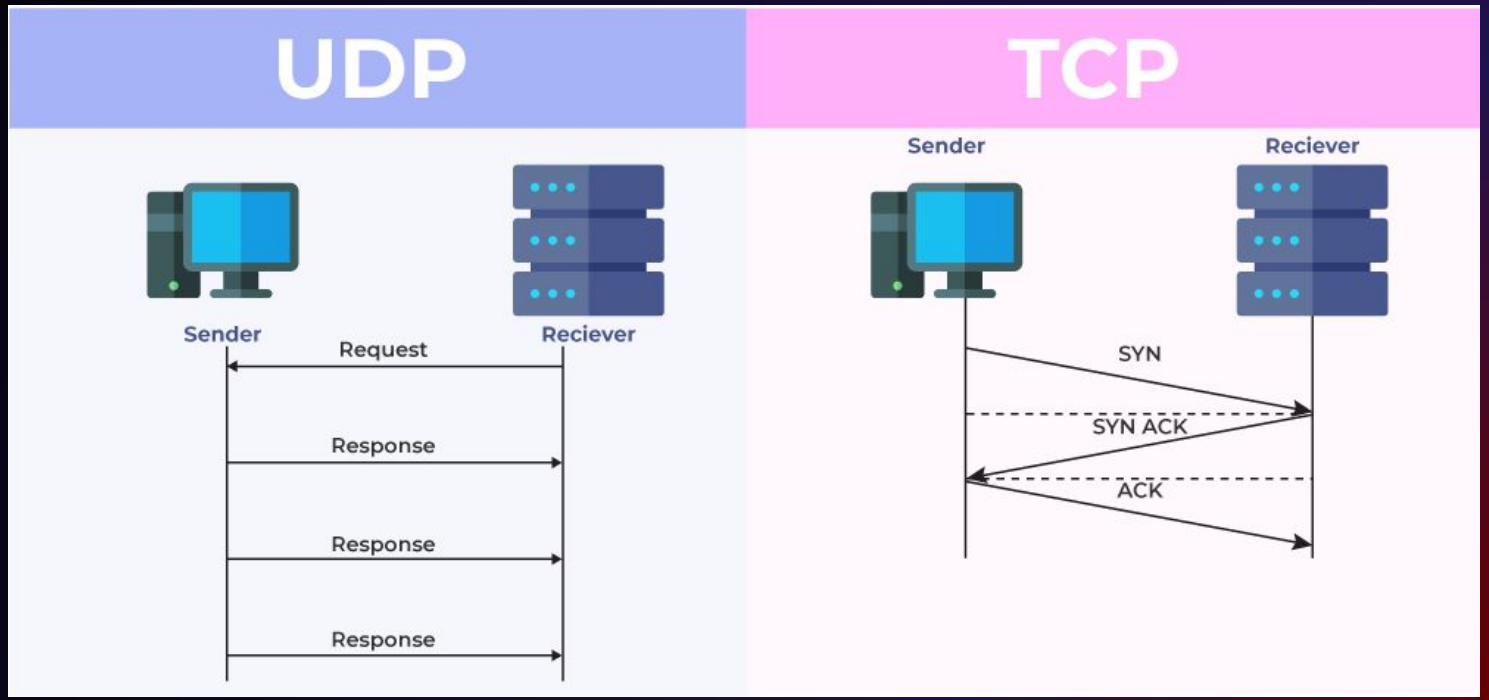
Two main protocols: TCP and UDP

TCP: makes a connection, checks data arrives, keeps correct order(file downloads)

UDP: faster, no guarantees, data can arrive late or out of order (video call)

Transport layer

TCP/IP model



Transport layer

TCP/IP model

Item	TCP	UDP
Stands For	Transmission Control Protocol	User Datagram Protocol
Protocol	Connection Oriented	Connectionless
Security	Makes Checks For Errors And Reporting	Makes Error Checking But No Reporting
Data Sending	Slower	Faster
Header Size	20 Bytes	8 Bytes
Segments	Acknowledgement	No Acknowledgement
Typical Applications	- Email	- VoIP

Application layer

TCP/IP model

The layer users and developers actually see. It defines how specific apps format and send their data

Example: HTTP and HTTPS load web pages

Example: DNS turns a name like google.com into an IP address

Example: SMTP sends email

Example: MQTT is used by IoT devices, we will use this later with the ESP32

Application layer

TCP/IP model

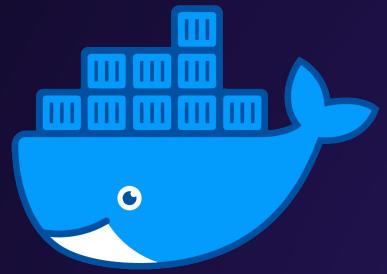




git







docker



POSTMAN

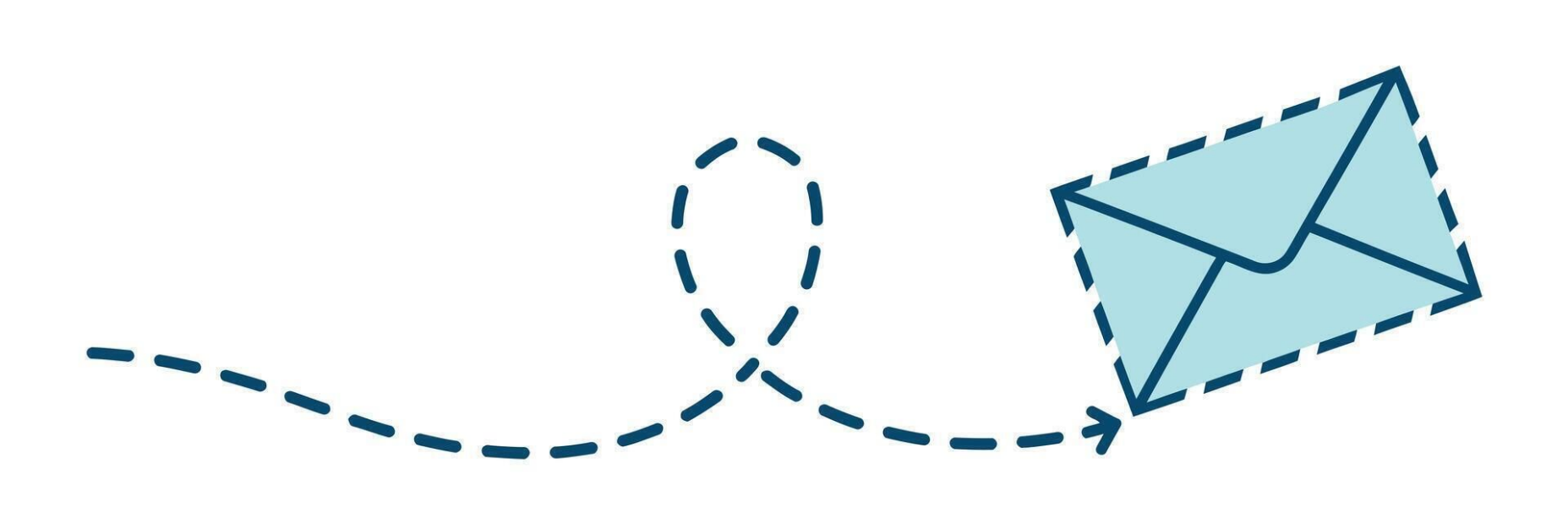
Encapsulation (message travel)

Data does not travel alone, each layer adds its own header before passing it down

On arrival, the receiving device removes each header in reverse order, this is called decapsulation

Same idea as the envelope: each layer adds its own wrapper, the layer below does not need to read what is inside

Encapsulation (message travel)



Encapsulation (message travel)

Application layer: creates the actual message

Transport layer: adds a header with port numbers, this turns the message into a segment

Network layer: adds a header with IP addresses, this turns the segment into a packet

Link layer: adds a header with MAC addresses, this turns the packet into a frame

Physical layer: sends the frame as raw bits

Wireshark software



Wireshark software

It is a free tool that captures real network traffic, packet by packet. It lets you see the exact layers we just covered: ethernet, IP, TCP, HTTP

https://www.wireshark.org/



Step by step



1 - Open Wireshark and select your interface. Pick the one showing live traffic (usually WiFi)

2 - Start the capture (click the shark-fin icon, traffic begins streaming immediately)

3 - Narrow the noise - type http in the filter bar so only plain web traffic shows

4 - Generate traffic - visit http://info.cern.ch , the world's first website, restored and still served over plain HTTP

Step by step



5 - Inspect a packet - click any row, then expand Frame → Ethernet → IP → TCP → HTTP in the panel below

6 - Match it to the model - MAC addresses at the Ethernet layer, IP addresses at the Network layer, port numbers at the Transport layer, the raw request at the Application layer

7 - Stop the capture - click the red square when done

Self test mcq

1. How many layers does the TCP/IP model have, compared to the OSI model's seven?

   A) 6 B) 3 C) 4 D) 7

2. At the Transport layer, a chunk of data being prepared for sending is called a:

   A) Segment B) Packet C) Frame D) Bit

Self test mcq

3. A device needs to decide the best path to forward data toward a destination on a different network. Which layer is responsible for this?

A) Physical B) Data Link C) Transport D) Network

4. You want to see only ICMP traffic (for example, from a ping command) in a live Wireshark capture. What do you type in the filter bar?

A) ping B) icmp C) http D) tcp.port==7

Self test mcq

5. Which statement best describes the real relationship between the OSI and TCP/IP models?

A) OSI survives today mainly as shared vocabulary and a troubleshooting framework

B) Both models were fully retired once cloud networking replaced older protocols

C) TCP/IP four layers map exactly one-to-one onto OSI seven layers

D) OSI is a literal blueprint for how the modern internet actually runs

Answers

1. C
2. A
3. D
4. B
5. A

Home work

Form your project team and draft an initial project topic. Come prepared to briefly justify your choice next class. What problem it solves and which of this semester's topics (software and hardware) it will draw on.